WELEARN GROWTH HUB | IT PLATFORM & ARCHITECTURE

Mastery Transcript Consortium (MTC) Integration

Technical Handover Specification, Data Mapping & API Integration Architecture

Version: 1.0.0-RC | Date: Nov 2026 | Author: Mani Golzary (Platform Lead) | Target: MTC & WeLearn Engineering

1. Executive Summary & Integration Purpose

Strategic alignment between WeLearn progressive learning and MTC Mastery Transcripts

WeLearn is a progressive mastery- and project-based academy operating across multiple cohorts, including secondary 2e learners (W2E), primary learners (WPR), and dual-enrolment students pursuing the Thai National Diploma. Our educational model rejects traditional seat-time and subjective letter grades (A-F), centering instead on verified competency demonstrations, iterative project rubrics, and rich multimodal learning evidence.

This technical handover document defines the bilateral integration between WeLearn Growth Hub (the school LMS and learning studio) and the Mastery Transcript Consortium (MTC) platform. The primary goal is to establish an automated, tamper-proof conduit that transmits coach-verified competency masteries and supporting student artifacts from WeLearn directly to MTC digital transcripts, empowering students with verified mastery credentials for university admissions and lifelong achievement.

CORE INTEGRATION OBJECTIVE

Enable continuous, verified transmission of student mastery milestones, rubric assessments, and authentic evidence URLs from WeLearn Growth Hub to MTC, with seamless single sign-on (SSO) and reciprocal credit framework synchronization.

2. LMS Environment & LTI Standards Baseline

Underlying learning platform infrastructure and specification compliance

WeLearn operates a custom-built, modern cloud learning platform--WeLearn Growth Hub--avoiding the constraints and legacy data models of commercial off-the-shelf LMS platforms.

- **Platform Core:** Single-Page React 18 client backed by Node.js / Express microservices hosted on Google Cloud Platform / Firebase (Enterprise Blaze).
- **Tenant Architecture:** Strict multi-tenancy enforcing cryptographic isolation between schools, cohorts (W2E, WPR, Thai Diploma), and classes.
- **1EdTech LTI 1.3 Advantage:** Engineered natively as an LTI 1.3 Tool Platform & Tool Consumer, fully certified for Core 1.3 (OIDC 3-legged launch), Assignment & Grade Services (AGS v2.0), Names and Role Provisioning Services (NRPS v2.0), and Deep Linking (v2.0).
- **Partner LMS Pass-Through:** For partner international schools maintaining Canvas LMS, WeLearn's backend acts as an authenticated LTI bridge, allowing upstream Canvas gradebook pass-through.

3. Security & Authentication Architecture

Dual-layered authentication for interactive sessions and server-to-server data pipelines

To satisfy international student privacy standards (FERPA, COPPA, and PDPA) while providing zero-friction usability, the integration operates two distinct, cryptographically isolated authentication planes:

3.1 Interactive Learner & Coach Launch (LTI 1.3 / OIDC)

When a student or coach opens their MTC portfolio viewer or launches WeLearn from within MTC, the session uses 1EdTech LTI 1.3 Advantage security based on OpenID Connect (OIDC) 3-legged redirection:

- **Step 1 (Login Initiation):**WeLearn initiates an OIDC authentication request to MTC's authorization endpoint with target_link_uri, lti_message_hint, and login_hint.
- **Step 2 (Authentication Response):**MTC authenticates the user context and returns a signed ID Token (JWT) via HTTP POST form submission to WeLearn's redirect URI.
- **Step 3 (JWKS Signature Verification):**WeLearn verifies the token signature against MTC's public JSON Web Key Set (/.well-known/jwks.json) using asymmetric RS256, validating state, nonce, and client_id.

3.2 Service-to-Service Automated REST API (OAuth 2.0)

All automated synchronization jobs--including event-driven webhook pushes and nightly reconciliation tasks--communicate with MTC's REST API via OAuth 2.0 Client Credentials Grant with asymmetric private key JWT authentication:

- **Client Token Exchange:** WeLearn signs an ephemeral assertion JWT with its private RSA key and POSTs to /oauth/token, receiving a scoped bearer access token (1 hour TTL).
- **API Authorization Header:** Subsequent REST API calls carry Authorization: Bearer <access_token> with tenant-scoping headers (X-School-ID, X-Cohort-ID).

DATA MINIMIZATION & COMPLIANCE RULE

Student personal support notes, psychological evaluations, and primary parent billing credentials never cross the API boundary. Only school-authorized academic mastery records, rubric evaluations, and public portfolio URLs are transmitted.

4. Bilateral Data Dictionary & Field Mapping

Complete cross-system mapping between WeLearn Growth Hub and MTC API schemas

WeLearn evaluates learning through four discrete rubric levels: Emerging (1), Developing (2), Proficient (3), and Exemplary (4). Mastery credit is earned when a learner achieves Proficient (3) or Exemplary (4) substantiated by coach-reviewed evidence artifacts.

WeLearn Object / Field	MTC API Field	Data Type	Mapping Rules & Transformation
learnerUid	`student.external_id`	UUID String	Immutable student ID; generated at enrollment.
`profile.displayName`	`student.first_name student.last_name`	String (split)	Parsed into given and family names; strips sensitive tags.
cohort	`student.cohort_id`	Enum String	Maps W2E (Senior), WPR (Primary), or Thai Diploma.
expectedGraduationYear	`student.grad_year`	Integer (YYYY)	Academic completion target (e.g., 2027).
competencyId	`credit.competency_id`	String / Code	Corresponds to MTC Credit Domain or custom STEAM standard.
rubricScore (3 or 4)	`evaluation.mastery_status`	Enum String	Score 3 -> 'PROFICIENT', Score 4 -> 'EXEMPLARY'.
evidenceUrl	`evidence.artifact_url`	HTTPS URI	Signed CDN URL to student submission (PDF, video, lab).
reflectionText	`evidence.narrative`	Markdown Text	Student's written meta-cognitive defense and summary.
coachUid	`evaluation.assessor_id`	UUID String	Immutable staff ID of the reviewing educator.
coachFeedback	`evaluation.coach_notes`	Markdown Text	Formal assessment notes and verification feedback.
evaluatedAt	`evaluation.awarded_at`	ISO 8601 UTC	Timestamp of coach signature (e.g., 2026-11-05T10:00:00Z).

5. Key API Endpoints & Request/Response Contracts

Technical payload contracts for core synchronization endpoints

POST `/api/v1/mastery-records`

Submits a coach-signed competency mastery achievement to MTC transcript queue.

- Header: Authorization: Bearer <access_token>
- Header: Content-Type: application/json
- Header: X-School-ID: welearn-bkk-01

```
{
  "student_external_id": "wl_stu_98b3f2e1",
  "competency_id": "MTC_SCI_ENERGY_04",
  "mastery_status": "PROFICIENT",
  "rubric_level": 3,
  "evaluated_at": "2026-11-05T08:30:00Z",
  "assessor": { "id": "wl_staff_mani_01", "name": "Mani Golzary", "role": "Coach" },
  "program_context": { "course_code": "STEAM-MICROGRID-26", "title": "Net-Zero Microgrid" }
}
```

POST `/api/v1/mastery-records/{record_id}/artifacts`

Attaches multimodal student evidence and reflective defense to an existing mastery record.

- Path Parameter: record_id (UUID returned from /mastery-records)
- Header: Idempotency-Key: <unique_hash_of_artifact>

```
{
  "artifact_type": "ENGINEERING_REPORT",
  "title": "Solar Energy Dispatch & SO2 Mitigation Modeling",
  "artifact_url": "https://storage.welearn.org/evidence/2026/wl_stu_98b3f2e1/microgrid_final.pdf",
  "media_format": "application/pdf",
  "file_size_bytes": 4829104,
  "student_narrative": "Synthesized solar car telemetry data to design a multi-zone battery buffer."
}
```

6. Data Flow Dynamics & Synchronization Cadence

Lifecycle of data exchange across triggers, frequencies, and directionality

To ensure transcript accuracy without placing unnecessary load on API infrastructure, data exchange operates under three distinct cadences:

Data Stream	Direction	Frequency	Operational Trigger & Resilience
Competency Framework	MTC -> WeLearn	Quarterly / On-Demand	Inbound pull to refresh standard MTC credit categories and competencies.
Roster & Provisioning	Two-way	Just-In-Time + Nightly	LTI launch auto-provisions learner; nightly batch syncs class status.
Mastery & Rubrics	WeLearn -> MTC	Real-Time Event	Fires within 5 seconds of coach signature; queues on network failure.
Multimodal Evidence	WeLearn -> MTC	Real-Time Event	Pushes verified HTTPS asset links immediately following rubric submission.
Transcript Publish Link	MTC -> WeLearn	On-Demand Read	Polled when parents or learners access the WeLearn Transcript Hub.

7. Open Technical Questions for Bilateral Meeting

Items for alignment during the scheduled working session with Jason and MTC

- **1. Sandbox Provisioning:** What is the procedure and lead time for provisioning a dedicated WeLearn sandbox tenant with mock student profiles?
- **2. Evidence Media Hosting:** Does MTC consume and cache raw media binaries (MP4/PDF up to 50MB), or does the MTC transcript viewer iframe/embed our secure HTTPS CDN URLs directly?
- **3. Revision & Resubmission Semantics:** If a learner improves their project after coach feedback and earns an upgrade from Proficient to Exemplary, what is MTC's preferred update pattern (PUT upsert vs. revision appending)?
- **4. Rate Limits & Concurrency:** What are the requests-per-minute rate limits, and is there a batch endpoint available for end-of-term transcript reconciliation (e.g., 500 records at once)?
- **5. Custom STEAM Competency Governance:** Can WeLearn register specialized NGSS/STEAM micro-competencies mapped parentally to standard MTC high-level credit areas?

8. Project Contacts & Bilateral Working Group

Key personnel and proposed meeting slots

- **MTC Lead:** Jason (Partner & Integration Lead)
- **WeLearn Platform Lead:** Mani Golzary (AI Learning Architect & IT Platform Lead, mani@welearn.org)
- **WeLearn Project Sponsor:** David Doran (Founder & CEO, WeLearn Academy)
- **Preferred Meeting Time:** 8:00 PM US Eastern Time (8:00 AM Indochina Time ICT) - Alternative: 6:00 AM US ET.

> **TECHNICAL READINESS CONFIRMATION**
>
> WeLearn's LTI 1.3 foundation and backend Express REST clients are ready for staging testing. Upon receipt of sandbox credentials from MTC, end-to-end integration tests can commence immediately within Phase B.